Domain-Specialized Genomic Language Models for Mitochondrial Genome Annotation: A Systematic Review

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Abstract

Background. Mitochondrial genome annotation remains a persistent computational challenge. Classical tools such as MITOS, MitoZ, and MFannot achieve high overall coverage but continue to fail systematically on transfer RNA (tRNA) gene prediction, gene boundary delineation in intron-containing genes, non-canonical genetic codes, and lineage-specific mitochondrial architectures. These failures propagate into variant interpretation pipelines, where annotation errors can directly affect pathogenicity calls for clinically significant mutations. In parallel, transformer-based genomic language models – including DNABERT-2, the Nucleotide Transformer, HyenaDNA, and Evo 2 – have demonstrated that treating DNA as a language enables extraction of long-range sequence patterns inaccessible to classical alignment or motif-based approaches. MitoGPT, the first language model fine-tuned specifically for mitochondrial genome annotation, builds upon DNABERT-2 and adapts it through a three-stage curriculum learning protocol to a suite of mitochondrial annotation tasks.

Objective. To what extent do domain-specialized genomic language models outperform general-purpose foundation models on organelle genome annotation, and what methodological and biological gaps remain before clinical deployment is feasible?

Methods. A systematic narrative review of the biological and data science literature underpinning mitochondrial genome annotation and genomic foundation models. Sources were identified through structured searches of PubMed, arXiv, bioRxiv, and Google Scholar. Inclusion criteria required peer-reviewed publication or registered preprint with direct relevance to one of twelve prespecified review domains. Twelve primary domains were reviewed across two broad categories: biological domain (five points) and data science domain (seven points).

Results. Existing annotation tools fail systematically on tRNA genes, intron-containing genes, and cross-species generalization. MitoGPT achieves 99.1% accuracy and macro-F1 of 0.979 on human mitochondrial gene type classification, validating the domain-specialization hypothesis for a three-class task. Key gaps identified include: absence of calibration analysis; single-species focus; unvalidated synthetic data augmentation; lack of multi-task evaluation; and no comparison against simpler baselines.

Conclusion. Domain-specialized fine-tuning of genomic foundation models on mitochondrial sequences is effective for basic annotation tasks but faces unresolved challenges in cross-species generalizability, evaluation methodology, and clinical-grade confidence estimation. Future work should prioritize calibration metrics, multi-task learning across all five proposed annotation tasks, and systematic cross-clade benchmarking.

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1 Introduction

1.1 Background and Rationale

Mitochondrial genomes occupy a unique position in genomic biology. Despite their compact size — encoding 37 genes within 16.6 kb in humans — they are disproportionately difficult to annotate automatically. Their architecture combines overlapping features, non-canonical codons, extensive regulatory signals, and remarkable structural diversity across species, all of which complicate automated annotation pipelines and necessitate expert intervention (4). The biological and clinical stakes are high: pathogenic variants in mitochondrial DNA are responsible for approximately 75% of adult-onset and 20–25% of pediatric-onset mitochondrial disease, estimated to affect at least 1 in 5,000 individuals (11).

In parallel, transformer-based language models have begun to revolutionize genomics. DNABERT-2 introduced byte-pair encoding (BPE) tokenization, multi-species pre-training, and architectural modifications including ALiBi attention, achieving dramatically improved computational efficiency while requiring 21× fewer parameters than competing models (14). More recently, Evo 2 extended genomic foundation modelling to 9 trillion base pairs spanning all domains of life, demonstrating that sufficiently large general models can predict the functional impact of genetic variants without task-specific fine-tuning (17).

Despite these advances, a dedicated mitochondrial language model had not been developed prior to MitoGPT. Existing genomic models are trained on whole genomes and do not exploit the unique structural regularities of mitochondrial DNA. MitoGPT was developed to address precisely this gap: it is the first language model fine-tuned specifically for mitochondrial genome annotation, building upon DNABERT-2 and adapting it through a curriculum-based protocol to increasingly complex mitochondrial tasks (**?**).

1.2 Review Question

This systematic review addresses the following primary question:

> *To what extent do domain-specialized genomic language models outperform general-purpose foundation models on organelle genome annotation, and what methodological and biological gaps remain before clinical deployment is feasible?*

This question is structured in PICO format as follows:

- **Population:** Mitochondrial DNA sequences across human and multi-species datasets

- **Intervention:** Domain-specialized fine-tuning of genomic language models (e.g., MitoGPT built on DNABERT-2)
- **Comparison:** General-purpose genomic foundation models (DNABERT-2, Nucleotide Transformer, HyenaDNA, Evo 2) and classical annotation tools (MITOS, MitoZ, MFannot)
- **Outcome:** Annotation accuracy (accuracy, macro-F1, MCC), calibration quality (ECE), cross-species generalizability, and clinical deployment readiness

2 Methods

2.1 Inclusion and Exclusion Criteria

2.1.1 Inclusion Criteria

- Peer-reviewed journal articles, conference proceedings (ICML, NeurIPS, ICLR), or registered preprints on arXiv/bioRxiv
- Direct relevance to at least one of twelve pre-specified review domains (Table 1)
- Published between 2008 and 2026 (with no lower bound for foundational methodological papers)
- Available in English
- Studies addressing DNA sequence modelling, mitochondrial biology, genomic annotation, transfer learning, tokenization, or model calibration

2.1.2 Exclusion Criteria

- Grey literature without peer review or registered preprint status (e.g., educational web pages, blog posts)
- Studies focused exclusively on protein or RNA sequences with no application to DNA
- Duplicate publications of the same study
- Papers where the full text was inaccessible after reasonable retrieval effort

2.2 Search Strategy

Structured searches were conducted across four databases: **PubMed**, **arXiv**, **bioRxiv**, and **Google Scholar**. Search terms were organized by domain and combined using Boolean operators. Primary search strings included:

- `mitochondrial genome annotation AND (MITOS OR MitoZ OR MFannot)`
- `genomic language model AND (DNABERT OR Nucleotide Transformer OR HyenaDNA OR Evo)`
- `DNA tokenization AND (BPE OR k-mer OR byte pair encoding)`
- `transfer learning AND genomics AND fine-tuning`
- `curriculum learning AND biological sequences`
- `synthetic data augmentation AND DNA sequences`
- `neural network calibration AND (ECE OR temperature scaling)`
- `multi-task learning AND genome annotation`
- `mitochondrial variant pathogenicity AND machine learning`
- `cross-species mitochondrial evolution AND Metazoa`
- `genomic database bias AND vertebrate overrepresentation`

Searches were conducted in June–July 2026. Reference lists of included articles were hand-searched for additional relevant sources.

2.3 Search Time Frame

The primary search covered publications from 2008 to July 2026. Foundational methodological papers (e.g., Bengio et al. 2009 on curriculum learning; Gissi et al. 2008 on Metazoan mitochondrial evolution) were included regardless of date given their direct conceptual relevance.

2.4 Level of Evidence

Sources were categorized using the following hierarchy:

1. Peer-reviewed systematic reviews and meta-analyses
2. Peer-reviewed original research articles in indexed journals
3. Peer-reviewed conference proceedings (ICML, NeurIPS, ICLR)
4. Registered preprints on arXiv or bioRxiv with subsequent peer-reviewed publication
5. Registered preprints without yet published peer-reviewed version

2.5 Quality Assessment

Each included source was assessed against the following criteria:

- Clarity and reproducibility of methodology
- Appropriateness of evaluation metrics for the claims made
- Presence of comparison to meaningful baselines
- Acknowledgement of limitations
- Relevance and specificity to the review domain

Studies used to support biological domain points were additionally assessed for taxonomic scope and generalizability of findings. Studies used to support data science domain points were additionally assessed for whether calibration and uncertainty metrics were reported alongside accuracy-based metrics.

2.6 Data Extraction and Synthesis

This review presents a **narrative synthesis** of findings, organized by twelve pre-specified domains across two categories (Table 1). Quantitative results from individual studies are reported as extracted from the original publications without statistical aggregation, as the heterogeneity of study designs and evaluation frameworks precludes meta-analysis.

Table 1: Twelve Pre-specified Review Domains

#	Category	Domain
1	Biological	Mitochondrial annotation: state of the art and failure modes
2	Biological	Biological specificity of mtDNA vs. nuclear DNA
3	Biological	Clinical relevance of annotation errors
4	Biological	Cross-species mitochondrial evolution and generalization
5	Biological	Dataset bias in mitochondrial genomics
6	Data Science	Genomic foundation models: comparative landscape
7	Data Science	Tokenization strategies for DNA sequences
8	Data Science	Transfer learning and domain adaptation in genomics
9	Data Science	Curriculum learning in biological sequence modelling
10	Data Science	Synthetic data generation for genomic tasks
11	Data Science	Evaluation methodology gaps
12	Data Science	Multi-task learning for genome annotation

3 Results

3.1 Biological Domain

3.1.1 Domain 1: Annotation Tools — State of the Art and Failure Modes

The automated annotation of mitochondrial genomes has evolved through successive tool generations, each addressing and partially inheriting the failures of its predecessors. MITOS was designed to compute consistent de novo annotations of metazoan mitogenomic sequences, achieving results matching RefSeq in annotation coverage and quality, yet even this improvement revealed systemic limitations: erroneous gene names, missing annotations, and particularly unreliable tRNA prediction (2). MitoZ extended this to a one-click pipeline integrating assembly, annotation, and visualization, recovering 97.33% of protein-coding genes (PCGs) and rRNA genes at high similarity — but with documented weaknesses in tRNA recovery and regulatory element detection (3).

The hardest annotation problems are structural rather than algorithmic. MFannot's critical analysis identified the most difficult annotation steps as: (i) structural modelling of intron-containing genes; (ii) identification and delineation of Group I and II introns; and (iii) identification of moderately conserved non-coding RNA genes (4). Additional challenges arise through genetic code evolution, which can redefine the translational identity of both start and stop codons, thus obscuring protein-coding genes. A large-scale comparative analysis of 2,831 vertebrate mitochondrial genomes confirmed a high number of errors in mitochondrial gene annotation, concentrated especially in tRNA genes (6).

DeGeCI (2023) further benchmarked multiple annotation tools in parallel, finding that no single tool wins across all gene types — each has systematic blind spots, and vertebrate genomes are consistently better annotated than invertebrate ones (5). The multi-failure profile of existing tools establishes the primary motivation for a learned, domain-specialized approach such as MitoGPT.

3.1.2 Domain 2: Biological Specificity of mtDNA vs. Nuclear DNA

The mitochondrial genome is architecturally distinct from the nuclear genome in ways that directly complicate computational annotation. Rackham & Filipovska (2022) provide the authoritative recent review of mitochondrial genome organization and expression, covering the tRNA punctuation model, D-loop architecture, strand asymmetry, and the unique gene expression mechanisms of mammalian mtDNA (7). Particularly important is the tRNA punctuation model: folded tRNAs act as secondary structure signals within polycistronic transcripts, such that annotation errors in tRNA genes cascade into errors in adjacent CDS boundaries — a compounding

failure mode absent in nuclear genomes.

Gustafsson, Falkenberg & Larsson (2016) established the biochemical foundation for understanding mtDNA expression machinery, demonstrating that mitochondria maintain a completely separate transcription system driven by a single-subunit RNA polymerase (POLRMT), with sequence signals in mtDNA promoters that are structurally unlike nuclear promoters (8). This explains why tools trained or designed for nuclear genome annotation systematically fail on mitochondrial sequences.

3.1.3 Domain 3: Clinical Relevance of Annotation Errors

Annotation quality is not an abstract bioinformatics concern. Of approximately 1,000 mutations currently associated with human diseases in MITOMAP, only approximately 10% have compelling evidence of being pathogenic based on published literature, while 86% remain unconfirmed (9). This vast grey zone is partly a consequence of annotation inconsistency: in the MT-ND1 gene alone, only 42 variants are reported in ClinVar whereas 693 are available in MITOMAP (10).

Standard nuclear DNA pathogenicity predictors — SIFT, PolyPhen-2, FatHmm, PROVEAN, MutationAssessor — perform poorly on mitochondrial variants because the underlying biology is structurally different (10). APOGEE 2, a domain-specific multi-layer ML model for mitochondrial missense variant interpretation, represents the closest published work to MitoGPT's `human_variant_cls` task, and demonstrates that domain-specialized models substantially outperform generic approaches (9). The persistence of 86% unconfirmed variants, and the diagnostic ambiguity this creates in clinical settings, directly motivates both improved annotation and improved variant classification pipelines.

A recent systematic analysis of 6,500 rare disease families found that the majority of candidate mitochondrial variants formally remain as variants of uncertain significance (VUS) according to ACMG/AMP guidelines (11). Additional evidence of pathogenicity requires sequencing additional tissues, segregation studies in maternal relatives, metabolite profiling, and functional studies — all resource-intensive steps that better computational annotation could help prioritize.

3.1.4 Domain 4: Cross-Species Mitochondrial Evolution and Model Generalizability

Mitochondrial genome architecture is far less conserved across the tree of life than vertebrate-centric studies imply. Gissi, Iannelli & Pesole (2008) reviewed structural features of mtDNA across more than 1,000 animal species, finding that the plasticity of Metazoan mtDNA is higher than previously thought and is mainly due to variation in number and location of tRNA genes (12). Vertebrates are unusual: their mtDNA shows exceptional stability in gene content, genome architecture, and gene

strand asymmetry, while invertebrates such as tunicates exhibit structural variability comparable to that found across non-deuterostome phyla.

This creates a fundamental generalizability problem for any model trained primarily on human sequences. The tRNA gene variation that drives cross-species structural differences is precisely the gene type where existing tools already fail. A model that achieves 99.1% accuracy on human mitochondrial gene type classification may fail substantially on invertebrate mitogenomes where the tRNA complement is rearranged. Furthermore, the annotation error rate in current NCBI databases varies dramatically by taxonomic class, with higher rearrangement rates observed in Actinopteri, Amphibia, and Reptilia (6). These results represent only currently available sequences, and increased representativeness in undersampled groups may reveal further divergence.

3.1.5 Domain 5: Dataset Bias in Mitochondrial Genomics

The genomic databases that train and evaluate mitochondrial models carry profound taxonomic and demographic biases. Hotaling, Kelley & Frandsen (2021) surveyed all 3,278 animal genome assemblies in GenBank and found that chordates account for 54% of all assemblies despite comprising only 3.9% of animal species; 34 times more assemblies are available for chordates relative to arthropods (13). Of 34 recognized animal phyla, 10 have zero genome sequences available.

Within human mitochondrial databases, bias operates at the population level. Bris et al. (2018) documented that Asian and African haplogroups represent only 21% and 13% respectively of the MITOMAP GenBank dataset, leading to potential misinterpretation of variant frequencies in non-European populations (10). MitoGPT's training corpus, despite integrating multiple public sources including MITOMAP, MITOBANK, NCBI Organelle, and MitoCarta 3.0, inherits these biases because it draws from the same public databases. The gold/silver/bronze tier system partially addresses data quality but does not address demographic or taxonomic underrepresentation.

3.2 Data Science Domain

3.2.1 Domain 6: Genomic Foundation Models — Comparative Landscape

The genomic foundation model landscape has evolved rapidly. DNABERT-2 (Zhou et al., 2023) replaced fixed k-mer tokenization with BPE, introduced multi-species pre-training, and achieved 21× fewer parameters than competing models while matching or exceeding them on most benchmarks (14). The Nucleotide Transformer (Dalla-Torre et al., 2025) scaled genomic models to 2.5 billion parameters across human and multi-species genomes, establishing standardized benchmarks (GUE) and

demonstrating that domain distance between pre-training and fine-tuning corpora critically affects downstream task performance (15). HyenaDNA (Nguyen et al., 2023) replaced attention with sub-quadratic implicit convolutions, enabling context lengths of up to 1 million nucleotides — 500 times longer than DNABERT-2 — at single nucleotide resolution (16).

Evo 2 (Brixi et al., 2026) represents the current frontier: a biological foundation model trained on 9 trillion DNA base pairs spanning all domains of life, capable of identifying disease-causing mutations and designing synthetic genomes without task-specific fine-tuning (17). Evo 2's existence raises a key question for MitoGPT's positioning: as general genomic models scale to all-of-life training corpora, does domain-specific fine-tuning remain necessary? MitoGPT's 99.1% accuracy demonstrates that fine-tuning is effective, but it does not yet demonstrate that it outperforms a zero-shot Evo 2 evaluation on the same task.

3.2.2 Domain 7: Tokenization Strategies for DNA Sequences

Tokenization represents a foundational modeling choice with direct biological consequences. Original DNABERT used fixed k-mers — splitting DNA into overlapping chunks of fixed size (e.g., always 6-mers) — which is computationally simple but biologically arbitrary, as functional boundaries in DNA do not respect fixed windows (14). DNABERT-2 switched to BPE, which learns which substrings appear most frequently and groups them accordingly. This allows the tokenizer to capture common biological patterns such as codon triplets and conserved motifs, and dramatically reduces vocabulary size while improving computational efficiency.

However, a BPE tokenizer trained on whole multi-species genomes may not optimally represent the specific sequence patterns of mitochondrial DNA. Mitochondria use non-canonical genetic codes, compact promoter architectures, and specific tRNA cloverleaf signals that could be more efficiently captured by a mitochondria-specific BPE vocabulary. Retraining the DNABERT-2 tokenizer on mtDNA-only corpora represents an untested experimental direction that could improve MitoGPT's sequence representation.

3.2.3 Domain 8: Transfer Learning and Domain Adaptation in Genomics

The Nucleotide Transformer benchmark established that pre-training on large genomic corpora helps on narrow downstream tasks, but gains are largest when pre-training data resembles the target task in terms of species and sequence type (15). When the target task is structurally distant from pre-training data, large models can perform no better than simpler baselines. DNABERT-2's pre-training corpus is dominated by nuclear sequences from diverse species; mitochondrial sequences constitute a

tiny fraction of this corpus. The amount of useful biological information actually transferred to MitoGPT's fine-tuning therefore remains an open question.

MitoGPT's rapid convergence — loss dropping from approximately 1.07 to below 0.01 within three epochs — suggests that the pre-trained weights provide a useful initialization. However, the near-ceiling accuracy (99.1%) on a three-class task may partially reflect that the task itself is easy given strong class-specific sequence signals (CDS vs. rRNA vs. tRNA), rather than that the DNABERT-2 backbone provides exceptional domain-specific representations.

3.2.4 Domain 9: Curriculum Learning in Biological Sequence Modelling

Curriculum learning was formalized by Bengio et al. (2009), who demonstrated that training on examples organized by increasing difficulty produces better generalization and faster convergence than random data ordering (18). The core insight is that starting with well-understood, high-quality examples and progressively introducing harder, noisier data guides the optimizer toward better local minima in non-convex loss landscapes.

MitoGPT implements a three-stage curriculum: Stage 1 uses gold-tier data (high-quality human annotations from rCRS GenBank features and confirmed MITOMAP disease variants); Stage 2 adds silver-tier data (cross-species GenBank features and MITOMAP polymorphisms); Stage 3 incorporates bronze-tier synthetic sequences. This design is theoretically justified by Bengio et al. (2009), and partially supported by multi-task genomics literature showing that staged training can improve task-specific performance (19). However, there is currently no direct empirical evidence that curriculum learning improves mitochondrial annotation specifically. The 99.1% accuracy reported by MitoGPT was achieved after Stage 1 training only, leaving the contribution of Stages 2 and 3 unmeasured. This is a key empirical gap.

3.2.5 Domain 10: Synthetic Data Generation for Genomic Tasks

MitoGPT generates 100,000 synthetic mitochondrial genomes using a `SyntheticGenerator` class that applies transition-biased random mutations (A↔G, C↔T) to the rCRS reference sequence. This approach is motivated by the relative scarcity of labeled mitochondrial sequence data.

Literature on synthetic data for regulatory genomics, however, urges caution. Bogard et al. (2024) found that random mutations, even when biologically inspired, can introduce patterns absent in real genomic sequences, and that evolutionary augmentation using real homologous sequences outperforms random mutation strategies for maintaining biological validity (20). A complementary review of augmentation methods for biological deep learning confirmed that traditional augmentation meth-

ods involving noise or irrelevant variations are often not feasible due to the risk of introducing biologically irrelevant changes (21).

MitoGPT's synthetic sequences are never validated against real biological constraints or used in ablation experiments to measure their specific contribution. Whether the 100k synthetic genomes help generalization or introduce spurious learned patterns remains an open empirical question.

3.2.6 Domain 11: Evaluation Methodology Gaps

MitoGPT reports accuracy (99.1%), macro-F1 (0.979), micro-F1 (0.991), and evaluation loss (0.043). These metrics are appropriate for measuring classification performance but are insufficient for clinical deployment readiness.

Guo, Pleiss, Sun & Weinberger (2017) established that modern deep neural networks are systematically overconfident: they produce high-confidence predictions (e.g., 90% probability) that are correct only at lower rates (e.g., 70%) (22). This miscalibration, measured by Expected Calibration Error (ECE), is particularly dangerous in clinical contexts where a model's confidence score may influence downstream diagnostic decisions. In variant pathogenicity assessment, a classifier that confidently misclassifies a pathogenic variant as benign is more harmful than one that correctly flags uncertainty.

MitoGPT reports no calibration analysis. Temperature scaling — a single-parameter post-hoc calibration method introduced by Guo et al. (2017) — is straightforward to implement and represents a minimal requirement before any clinical application. The absence of calibration evaluation in MitoGPT is the single most actionable methodological gap identified in this review.

Additionally, MitoGPT does not compare against simpler baselines (k-mer logistic regression, CNN trained from scratch, classical tools on the same evaluation split). Without such comparisons, it is not possible to attribute the 99.1% accuracy specifically to the DNABERT-2 backbone rather than to the inherent simplicity of the three-class task.

3.2.7 Domain 12: Multi-Task Learning for Genome Annotation

MitoGPT proposes a shared DNABERT-2 encoder with task-specific classification heads for five annotation tasks: gene type classification, gene boundary detection, regulatory region annotation, variant classification, and cross-species gene type classification. This design is theoretically motivated by evidence that shared encoders with task-specific heads improve performance across related genomic tasks (19), and by broader multi-task biology literature showing that tasks sharing biological structure benefit from joint representation learning (19).

However, the five MitoGPT tasks span a wide range of biological complexity and annotation granularity. Gene type classification (CDS, rRNA, tRNA) involves relatively coarse sequence-level patterns strongly represented in the pre-trained DNABERT-2 backbone. Variant pathogenicity scoring requires fine-grained interpretation of single-nucleotide changes in a clinical context. Whether these tasks share enough learned representation structure to benefit from a shared encoder — or whether conflicting gradient signals degrade performance on individual tasks — is an untested assumption. The current MitoGPT publication evaluates only the gene type task; four of the five proposed tasks remain unimplemented and unevaluated.

4 Discussion

4.1 Summary of Main Results

This review synthesizes 22 primary sources across twelve domains relevant to MitoGPT's design and evaluation. Across both the biological and data science domains, a consistent picture emerges: MitoGPT represents a well-motivated proof-of-concept that succeeds at demonstrating the feasibility of domain-specialized genomic language model fine-tuning, but faces important unresolved challenges before it can be considered a complete mitochondrial annotation system.

On the biological side, the three core failures of existing annotation tools — tRNA prediction, cross-species generalization, and regulatory element detection — are well documented and constitute a clear opportunity for a learned model. MitoGPT's strong performance on gene type classification validates that DNABERT-2 pre-training transfers usefully to the mitochondrial domain. However, the task evaluated (three-class CDS/rRNA/tRNA classification on human sequences) is the easiest of the five proposed MitoGPT tasks and the one least likely to surface the model's failure modes.

On the data science side, the most significant finding is the absence of calibration analysis. In clinical variant interpretation, confidence scores are as important as raw accuracy. The systematic overconfidence of modern neural networks documented by Guo et al. (2017) is a known problem that has not been addressed in MitoGPT.

4.2 Comparison with Existing Literature

MitoGPT's 99.1% accuracy and 0.979 macro-F1 on gene type classification substantially exceeds the GUE benchmark range (0.3–0.8 F1 across various genomic tasks), suggesting that mitochondrial gene type classification is a well-posed problem with strong discriminative signals (14). However, the comparison is difficult because MitoGPT uses a custom evaluation set rather than a shared benchmark.

The closest related work is APOGEE 2 (9), which addresses mitochondrial variant pathogenicity using a multi-layer ML model. APOGEE 2 demonstrates that domain-specific models outperform generic nuclear variant predictors on mitochondrial variants, aligning with MitoGPT's central hypothesis. Neither APOGEE 2 nor any prior work has attempted multi-task mitochondrial annotation combining gene type, boundary, regulatory, and variant tasks in a single model.

Evo 2 (17) represents the most important emerging challenge to MitoGPT's positioning. If a sufficiently large general genomic model trained on all domains of life can accurately annotate mitochondrial sequences without fine-tuning, the marginal value of domain-specialized models decreases. Whether this is the case for the full MitoGPT task suite — particularly for the clinically sensitive variant classification task — remains an open and pressing research question.

4.3 Limitations of This Review

1. **Scope:** This review focuses on a single model (MitoGPT) and its supporting literature. A broader systematic comparison of all genomic foundation models on organelle tasks is beyond its scope.

2. **Heterogeneity:** The twelve domains span molecular biology, clinical genomics, and machine learning. Evaluation standards differ substantially across these communities, preventing quantitative aggregation.

3. **Preprint inclusion:** Several key sources (HyenaDNA, MitoGPT) were initially published as preprints. Where peer-reviewed versions exist, these are cited; where not, the preprint is cited with explicit note.

4. **Knowledge cutoff:** This review reflects the literature available as of July 2026. The genomic foundation model field is advancing rapidly; results from models released after Evo 2 may alter the landscape.

5. **Narrative synthesis:** The heterogeneity of included studies precludes meta-analysis. All quantitative comparisons are narrative.

4.4 Implications for Policy, Practice and Future Research

Immediate methodological priorities. MitoGPT should add calibration analysis (ECE, reliability diagrams, temperature scaling) before any clinical pilot. This is low-cost and high-impact. Comparison against k-mer logistic regression and CNN baselines on the same evaluation split would also clarify how much performance is attributable to the DNABERT-2 backbone vs. task simplicity.

Near-term research priorities. Evaluation of the remaining four MitoGPT tasks (`human_regulatory`, `human_gene_boundary`, `human_variant_cls`, `multispecies_gene_type`) is essential to validate the multi-task hypothesis. Cross-species evaluation on invertebrate mitogenomes — particularly arthropods, molluscs, and annelids — will reveal whether the model's performance reflects genuine mitochondrial biology or vertebrate-centric training signal.

Long-term research priorities. Development of a mitochondria-specific BPE tokenizer retrained on mtDNA-only corpora could improve sequence representation. Integration of calibrated MitoGPT predictions into existing variant annotation pipelines (MITOMAP, MSeqDR) would provide the first direct measurement of clinical utility.

5 Conclusion

Domain-specialized fine-tuning of genomic language models for mitochondrial annotation is both feasible and effective for basic classification tasks. MitoGPT achieves near-ceiling performance (99.1% accuracy, 0.979 macro-F1) on human mitochondrial gene type classification using a three-stage curriculum learning protocol applied to a curated multi-source dataset, validating the core hypothesis that domain-specific training improves over general genomic models for mitochondrial annotation tasks.

However, the current results address only the easiest of five proposed annotation tasks and focus exclusively on human sequences. The biological literature documents clear failure modes — particularly for tRNA annotation across species — that the simple three-class evaluation does not expose. The data science literature identifies calibration as a critical missing evaluation dimension, and raises unresolved questions about the value of synthetic data augmentation and the conditions under which multi-task learning helps or interferes.

The field of genomic foundation models is advancing rapidly. Evo 2's demonstration that a sufficiently large all-of-life model can predict variant effects without fine-tuning challenges the long-term positioning of domain-specialized approaches. The value of domain specialization will ultimately be determined by performance on the tasks where general models are weakest: cross-species annotation of structurally divergent invertebrate mitogenomes, fine-grained regulatory element annotation, and clinically calibrated variant pathogenicity prediction.

Future work should prioritize: calibration analysis; multi-task evaluation across all five proposed tasks; cross-clade benchmarking; systematic baseline comparison; and eventual integration into clinical variant annotation workflows. MitoGPT provides a solid technical foundation for this research agenda.

References

[1] National Collaborating Centre for Methods and Tools (2014). *Anatomy of a Systematic Review* [fact sheet]. Retrieved from http://www.nccmt.ca

[2] Bernt, M., Donath, A., Jühling, F., Externbrink, F., Florentz, C., Fritzsch, G., Pütz, J., Middendorf, M., & Stadler, P. F. (2013). MITOS: Improved de novo metazoan mitochondrial genome annotation. *Molecular Phylogenetics and Evolution*, 69(2), 313–319. https://doi.org/10.1016/j.ympev.2012.08.023

[3] Meng, G., Li, Y., Yang, C., & Liu, S. (2019). MitoZ: A toolkit for animal mitochondrial genome assembly, annotation and visualization. *Nucleic Acids Research*, 47(11), e63. https://doi.org/10.1093/nar/gkz173

[4] Lang, B. F., Beck, N., Prince, S., Sarrasin, M., Rioux, P., & Burger, G. (2023). Mitochondrial genome annotation with MFannot: a critical analysis of gene identification and gene model prediction. *Frontiers in Plant Science*, 14, 1222186. https://doi.org/10.3389/fpls.2023.1222186

[5] Fiedler, J., et al. (2023). DeGeCI: Fully automated annotation of mitochondrial genomes using a cluster-based approach with de Bruijn graphs. *Frontiers in Genetics*, 14, 1250907. https://doi.org/10.3389/fgene.2023.1250907

[6] Montaña-Lozano, P., Moreno-Carmona, M., Ochoa-Capera, M., Medina, N. S., & Boore, J. L. (2022). Comparative genomic analysis of vertebrate mitochondrial reveals a differential of rearrangements rate between taxonomic class. *Scientific Reports*, 12, 5479. https://doi.org/10.1038/s41598-022-09512-2

[7] Rackham, O., & Filipovska, A. (2022). Organization and expression of the mammalian mitochondrial genome. *Nature Reviews Genetics*, 23(10), 606–623. https://doi.org/10.1038/s41576-022-00480-x

[8] Gustafsson, C. M., Falkenberg, M., & Larsson, N.-G. (2016). Maintenance and expression of mammalian mitochondrial DNA. *Annual Review of Biochemistry*, 85, 133–160. https://doi.org/10.1146/annurev-biochem-060815-014402

[9] Bianco, S. D., Parca, L., Petrizzelli, F., Biagini, T., Giovannetti, A., Liorni, N., Napoli, A., Carella, M., Procaccio, V., Lott, M. T., Zhang, S., Vescovi, A. L., Wallace, D. C., Caputo, V., & Mazza, T. (2023). APOGEE 2: Multi-layer machine-learning model for the interpretable prediction of mitochondrial missense variants. *Nature Communications*, 14, 5058. https://doi.org/10.1038/s41467-023-40797-7

[10] Bris, C., et al. (2018). Bioinformatics tools and databases to assess the pathogenicity of mitochondrial DNA variants in the field of next generation sequencing. *Frontiers in Genetics*, 9, 632. https://doi.org/10.3389/fgene.2018.00632

[11] Variants detected in 6,500 rare disease families by systematic analysis of exome and genome sequencing (2025). *HGG Advances*. https://doi.org/10.1016/j.xhgg.2025.100439

[12] Gissi, C., Iannelli, F., & Pesole, G. (2008). Evolution of the mitochondrial genome of Metazoa as exemplified by comparison of congeneric species. *Heredity*, 101(4), 301–320. https://doi.org/10.1038/hdy.2008.62

[13] Hotaling, S., Kelley, J. L., & Frandsen, P. B. (2021). Toward a genome sequence for every animal: Where are we now? *Proceedings of the National Academy of Sciences*, 118(49), e2109019118. https://doi.org/10.1073/pnas.2109019118

[14] Zhou, Z., Ji, Y., Li, W., Dutta, P., Davuluri, R. V., & Liu, H. (2023). DNABERT-2: Efficient foundation model and benchmark for multi-species genome. *arXiv preprint* arXiv:2306.15006. https://arxiv.org/abs/2306.15006

[15] Dalla-Torre, H., et al. (2025). Nucleotide Transformer: Building and evaluating robust foundation models for human genomics. *Nature Methods*, 22(2), 287–297. https://doi.org/10.1038/s41592-024-02523-z

[16] Nguyen, E., Poli, M., Faizi, M., Thomas, A. W., Birch-Sykes, C., Wornow, M., Patel, A., Rabideau, C., Massaroli, S., Bengio, Y., Ermon, S., Baccus, S. A., & Ré, C. (2023). HyenaDNA: Long-range genomic sequence modeling at single nucleotide resolution. *arXiv preprint* arXiv:2306.15794. https://arxiv.org/abs/2306.15794

[17] Brixi, G., Durrant, M. G., Ku, J., et al. (2026). Genome modelling and design across all domains of life with Evo 2. *Nature*, 652, 1349–1361. https://doi.org/10.1038/s41586-026-10176-5

[18] Bengio, Y., Louradour, J., Collobert, R., & Weston, J. (2009). Curriculum learning. *Proceedings of the 26th Annual International Conference on Machine Learning (ICML '09)*, 41–48. ACM Press. https://doi.org/10.1145/1553374.1553380

[19] Badré, A., et al. (2023). Explainable multi-task learning improves the parallel estimation of polygenic risk scores for many diseases through shared genetic basis. *PLOS Computational Biology*, 19(7), e1011211. https://doi.org/10.1371/journal.pcbi.1011211

[20] Bogard, B., et al. (2024). Improving the performance of supervised deep learning for regulatory genomics using phylogenetic augmentation. *Bioinformatics*, 40(4), btae190. https://doi.org/10.1093/bioinformatics/btae190

[21] Innovative data augmentation strategy for deep learning on biological datasets with limited gene representations focused on chloroplast genomes (2025). *Scientific Reports*. https://doi.org/10.1038/s41598-025-12796-9

[22] Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. *Proceedings of the 34th International Conference on Machine Learning (ICML 2017)*, 70, 1321–1330. https://arxiv.org/abs/1706.04599